

AI Swarm Drones Find Survivors 3-4x Faster Than Traditional SAR

10-18 MIN/MI²78-85% DETECTION12% FALSE ALARMS₹95,000 SORTIE



50Hz OFFBOARD

SUBSYSTEM A // GNC

SUTRA-FSD GNC & 3D ORCA

SOLVES PROBLEM VOID:

Canopy GPS Multi-Path Drift & Crashes

SOLUTIONAL MOAT:

Decentralized 3D velocity obstacle collision avoidance + PX4 MicroXRCE 50Hz offboard setpoint streaming.

0.05s

Response Latency

0 Tower

Dependency

DECENTRALIZED VIO

50Hz RATE



-5dB RESILIENT

SUBSYSTEM B // COMMS

Deep JSCC Neural Video

SOLVES PROBLEM VOID:

Mountain Ravine RF Ridge Blackout

SOLUTIONAL MOAT:

Differentiable joint source-channel coding yielding zero digital cliff and graceful video degradation down to -5dB.

-5dB

SNR Threshold

0 Cliff

Degradation

SEMANTIC COMMS

EDGE TPU



3.59cm ACCURACY

SUBSYSTEM C // VISION

3D DEM AI Geolocation

SOLVES PROBLEM VOID:

35% False Alarms & Flat GPS Drift

SOLUTIONAL MOAT:

Sub-15ms edge YOLOv8-Pose with real-time WGS84 elevation raycasting, eliminating 15-30m calculation errors.

<15ms

Inference Time

3.59cm

Geo Error

ELEVATION CORRECTED

JETSON ORIN



1-2 OPERATORS

SUBSYSTEM D // C2 GCS

Pegasus 3D WebGPU GCS

SOLVES PROBLEM VOID:

15-25 Ground Crew Cognitive Overload

SOLUTIONAL MOAT:

Offline tactical 3D digital twin + ATAK plugin, empowering 1-2 operators to orchestrate full drone swarms.

1-2 Ops

Crew Size

100%

Offline ATAK

ATAK INTEGRATION

WEBGPU 3D